

5.1 The correct matching for the following pairs is

List-I	List-II
A. DMA I/O	1. High speed RAM
B. Cache	2. Disk
C. Interrupt I/O	3. Printer
D. Condition Code Register	4. ALU

Codes:

- (a) A-4 B-3 C-1 D-2 (b) A-2 B-1 C-3 D-4
(c) A-4 B-3 C-2 D-1 (d) A-2 B-3 C-4 D-1

[1997 : 1 M]

5.2 The correct matching for the following pairs is

List-I	List-I
A. Disk scheduling	1. Round robin
B. Batch processing	2. SCAN
C. Time sharing	3. LIFO
D. Interrupt processing	4. FIFO

Codes:

- (a) A-3, B-4, C-2, D-1
(b) A-4, B-3, C-2, D-1
(c) A-2, B-4, C-1, D-3
(d) A-3, B-4, C-3, D-2

[1997 : 1 M]

5.3 I/O redirection

- (a) implies changing the name of a file
(b) can be employed to use an existing file as input file for a program
(c) implies connecting 2 programs through a pipe
(d) None of the above

[1997 : 1 M]

5.4 When an interrupt occurs, an operating system

- (a) ignores the interrupt
(b) always changes state of interrupted process after processing the interrupt
(c) always resumes execution of interrupted process after processing the interrupt
(d) may change state of interrupted process to 'blocked' and schedule another process

[1997 : 1 M]

5.5 Which of the following devices should get higher priority in assigning interrupts?

- (a) Hard disk (b) Printer
(c) Keyboard (d) Floppy disk

[1998 : 1 M]

5.6 Which of the following is true?

- (a) Unless enabled, a CPU will not be able to process interrupts.
(b) Loop instructions cannot be interrupted till they complete.
(c) A processor checks for interrupts before executing a new instruction.
(d) Only level triggered interrupts are possible on microprocessors

[1998 : 1 M]

5.7 Listed below are some operating system abstractions (in the left column) and the hardware components (in the right column)?

A. Thread	1. Interrupt
B. Virtual address space	2. Memory
C. File system	3. CPU
D. Signal	4. Disk

- (a) A-2, B-4, C-3, D-1
(b) A-1, B-2, C-3, D-4
(c) A-3, B-2, C-4, D-1
(d) A-4, B-1, C-2, D-3

[1999 : 1 M]

5.8 Which of the following disk scheduling strategies is likely to give the best throughput?

- (a) Farthest cylinder next
(b) Nearest cylinder next
(c) First come first served
(d) Elevator algorithm

[1999 : 1 M]

5.9 Which of the following requires a device driver?

- (a) Register (b) Cache
(c) Main memory (d) Disk

[2001 : 1 M]

5.10 In the index allocation scheme of blocks to a file, the maximum possible size of the file depends on

- (a) the size of the blocks, and the size of the address of the blocks.
(b) the number of blocks used for the index, and the size of the blocks
(c) the size of the blocks, the number of blocks used for the index, and the size of the address of the blocks
(d) None of the above

[2002 : 2 M]

5.11 Using a larger block size in a fixed block size file system leads to

- (a) better disk throughput but poorer disk space utilization

- (b) better disk throughput and better disk space utilization
- (c) poorer disk throughput but better disk space utilization
- (d) poorer disk throughput and poorer disk space utilization

[2003 : 1 M]

5.12 A Unix-style I-node has 10 direct pointers and one single, one double and one triple indirect pointers. Disk block size is 1 Kbyte, disk block address is 32 bits, and 48-bit integers are used. What is the maximum possible file size?

- (a) 2^{24} bytes
- (b) 2^{32} bytes
- (c) 2^{34} bytes
- (d) 2^{48} bytes

[2004 : 2 M]

5.13 A disk has 200 tracks (numbered 0 through 199). At a given time, it was servicing the request of reading data from track 120, and at the previous request, service was for track 90. The pending requests (in order of their arrival) are for track numbers.

30 70 115 130 110 80 20 25

How many times will the head change its direction for the disk scheduling policies SSTF (Shortest Seek Time First) and FCFS (First Come First Serve)?

- (a) 2 and 3
- (b) 3 and 3
- (c) 3 and 4
- (d) 4 and 4

[2004 : 2 M]

5.14 In a particular Unix OS, each data block is of size 1024 bytes, each node has 10 direct data block addresses and three additional addresses: one for single indirect block, one for double indirect block and one for triple indirect block. Also, each block can contain addresses for 128 blocks. Which one of the following is approximately the maximum size of a file in the file system?

- (a) 512 MB
- (b) 2 GB
- (c) 8 GB
- (d) 16 GB

[2004 : 2 M]

5.15 In a computer system, four files of size 11050 bytes, 4990 bytes, 5170 bytes and 12640 bytes need to be stored. For storing these files on disk, we can use either 100 byte disk blocks or 200 byte disk blocks (but can't mix block sizes). For each block used to store a file, 4 bytes of bookkeeping information also needs to be stored on the disk. Thus, the total space used to store a file is the sum of the space taken to store the file and the space taken to store the book keeping information for the blocks allocated for storing

the file. A disk block can store either bookkeeping information for a file or data from a file, but not both.

What is the total space required for storing the files using 100 byte disk blocks and 200 byte disk blocks respectively?

- (a) 35400 and 35800 bytes
- (b) 35800 and 35400 bytes
- (c) 35600 and 35400 bytes
- (d) 35400 and 35600 bytes

[2005 : 2 M]

5.16 In the working-set strategy, which of the following is done by the operating system to prevent thrashing?

- I. It initiates another process if there are enough extra frames.
- II. It selects a process to suspend if the sum of the sizes of the working-sets.
- III. Exceeds the total number of available frames.

- (a) I only
- (b) II only
- (c) Neither I nor II
- (d) Both I and II

[2006 : 1 M]

Directions for Question 5.17 to 5.18:

The head of a hard disk serves requests following the shortest seek time first (SSTF) policy. The head is initially positioned at track number 180.

5.17 Which of the request sets will cause the head to change its direction after servicing every request assuming that the head does not change direction if there is a tie in SSTF and all the requests arrive before the servicing starts?

- (a) 11, 139, 170, 178, 181, 184, 201, 265
- (b) 10, 138, 170, 178, 181, 185, 201, 265
- (c) 10, 139, 169, 178, 181, 184, 201, 265
- (d) 10, 138, 170, 178, 181, 185, 200, 265

[2007 : 2 M]

5.18 What is the maximum cardinality of the request set, so that the head changes its direction after servicing every request if the total number of tracks are 2048 and the head can start from any track?

- (a) 9
- (b) 10
- (c) 11
- (d) 12

[2007 : 2 M]

5.19 A virtual memory system uses First In First Out (FIFO) page replacement policy and allocates a fixed number of frames to a process. Consider the following statements:

P : Increasing the number of page frames allocated to a process sometimes increases the page fault rate.

Q : Some programs do not exhibit locality of reference.

Which one of the following is TRUE?

- (a) Both P and Q are true, and Q is the reason for P
- (b) Both P and Q are true, but Q is not the reason for P
- (c) P is false, but Q is true
- (d) Both P and Q are false

[2007 : 2 M]

5.20 The data block of a very large file in the Unix file system are allocated using

- (a) Contiguous allocation
- (b) Linked allocation
- (c) indexed allocation
- (d) an extension of indexed allocation

[2008 : 1 M]

5.21 Consider a disk system with 100 cylinders. The requests to access the cylinders occur in following sequence:

4, 34, 10, 7, 19, 73, 2, 15, 6, 20

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and shortest seek time first policy is used?

- (a) 95 ms
- (b) 119 ms
- (c) 233 ms
- (d) 276 ms

[2009 : 2 M]

5.22 A multilevel page table is preferred in comparison to a single level page table for translating virtual address to physical address because

- (a) it reduces the memory access time to read or write a memory location
- (b) it helps to reduce the size of page table needed to implement the virtual address space of a process
- (c) it is required by the translation look a side buffer
- (d) it helps to reduce the number of page faults in page replacement algorithms.

[2009 : 2 M]

5.23 A file system with 300 GByte disk uses a file descriptor with 8 direct block addresses, 1 indirect block address and 1 doubly indirect block address. The size of each disk block is 128 Bytes and the size of each disk block address is 8 Bytes. The maximum possible file size in this file system is

- (a) 3 KBytes
- (b) 35 KBytes
- (c) 280 KBytes
- (d) dependent on the size of the disk

[2012 : 2 M]

5.24 Suppose a disk has 201 cylinders, numbered from 0 to 200. At some time the disk arm is at cylinder 100, and there is a queue of disk access requests for cylinders 30, 85, 90, 100, 105, 110, 135 and 145. If Shortest-Seek Time First (SSTF) is being used for scheduling the disk access, the request for cylinder 90 is serviced after servicing _____ number of requests. [2014 (Set-1) : 1 M]

5.25 A FAT (file allocation table) based file system is being used and the total overhead of each entry in the FAT is 4 bytes in size. Given a 100×10^6 bytes disk on which the file system is stored and data block size is 10^3 bytes, the maximum size of a file that can be stored on this disk in units of 10^6 bytes is _____. [2014 (Set-2) : 1 M]

5.26 Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, 25, 70. Assume that the initial position of the R/W head is on track 50. The additional distance that will be traversed by the R/W head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves towards 100 when it starts execution) is _____ tracks. [2015 (Set-1) : 2 M]

5.27 Consider a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is _____. [2016 (Set-1) : 2 M]

5.28 In a file allocation system, which of the following allocation scheme(s) can be used if no external fragmentation is allowed?

- I. Contiguous
- II. Linked
- III. Indexed

- (a) I and III only
- (b) II only
- (c) III only
- (d) II and III only

[2017 (Set-2) : 1 M]

5.29 Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ..., 199) and 256 sectors per track (numbered as 0, 1, ..., 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2], [180, 134, 1], [60, 20, 0],

[212, 86, 3], [56, 116, 2], [118, 16, 1]

Currently the head is positioned at sector number 100 of cylinder 80 and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible.

The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____. [2018 : 2 M]

530 The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointers. The disk block size is 4 kB and the disk block address is 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) _____ GB. [2019 : 2 M]

531 Consider the following five disk access requests of the form (request id, cylinder number) that are present in the disk scheduler queue at a given time.

(P, 155), (Q, 85), (R, 110), (S, 30), (T, 115)

Assume the head is positioned at cylinder 100. The scheduler follows Shortest Seek Time First scheduling to service the requests,

Which one of the following statements is FALSE?

- (a) R is serviced before P.
- (b) T is serviced before P.
- (c) Q is serviced after S but before T.
- (d) The head reverses its direction of movement between servicing of Q and P. [2020 : 2 M]

532 Consider a linear list based directory implementation in a file system. Each directory is a list of nodes, where each node contains the file name along with the file metadata, such as the list of pointers to the data blocks. Consider a given directory foo.

Which of the following operations will necessarily require a full scan of foo for successful completion?

- (a) Opening of an existing file in foo
- (b) Renaming of an existing file in foo
- (c) Creation of a new file in foo
- (d) Deletion of an existing file from foo [2021 (Set-1) : 1 M]

533 Consider two files systems A and B, that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already

stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B , respectively, then the value of $n_A + n_B$ is _____. [2022 : 2 M]

5.34 Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each sector holds 1024 bytes of data. The number of cylinders in the hard disk is _____. [2024 (Set-1) : 2 M]

5.35 Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512 byte sectors. A file has content stored in 3000 sectors located randomly on the disk. Assuming average rotational latency, the total time (in seconds, rounded off to 2 decimal places) to read the entire file from the disk is _____. [2024 (Set-2) : 2 M]

5.36 A disk of size 512 M bytes is divided into blocks of 64 K bytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a NULL pointer (also of 4 bytes). Suppose a file of 1 M bytes needs to be stored in the disk. Assume, $1 \text{ K} = 2^{10}$ and $1 \text{ M} = 2^{20}$. The amount of space in bytes that will be wasted due to internal fragmentation is _____. (Answer in integer) [2025 (Set-1) : 2 M]

5.37 To keep track of free blocks in a file system, one of the two approaches is generally used – using bitmaps (bit vectors) or using linked lists. Consider that the linked list approach is used to keep track of free blocks in a file system. Assume that the disk size is 16 GB, block size is 2 KB, and block numbers used are 32-bit long. A single pointer of size 4 bytes is used in each block of the list to point to the next block of the list. The number of blocks required to hold the free disk block numbers is _____. (answer in integer) [2026 (Set-2) : 2 M]

Note: $1 \text{ K} = 2^{10}$ and $1 \text{ G} = 2^{30}$

Answers File System and Device Management

5.1 (b)	5.2 (c)	5.3 (b)	5.4 (d)	5.5 (a)	5.6 (a, c)	5.7 (c)	5.8 (b)	5.9 (d)
5.10 (c)	5.11 (a)	5.12 (c)	5.13 (c)	5.14 (b)	5.15 (c)	5.16 (b)	5.17 (b)	5.18 (c)
5.19 (b)	5.20 (d)	5.21 (b)	5.22 (b)	5.23 (b)	5.24 (3)	5.25 (99.6)	5.26 (10)	5.27 (346)
5.28 (d)	5.29 (85)	5.30 (4)	5.31 (c)	5.32 (b, c)	5.33 (153)	5.34 (4096)	5.35 (30.06)	
5.36 (65468)	5.37 (16417)							

Explanations File System and Device Management**5.1 (b)**

DMA I/O → Disk

Cache → High speed RAM

Interrupt I/O → Printer

Condition code register → ALU

5.2 (c)

Disk scheduling → SCAN

Batch processing → FIFO (Queue)

Time sharing → Round Robin (uses time slice)

Interrupt processing → LIFO (stack)

5.3 (b)

I/O redirection helps to use an existing file as input file for a program.

5.4 (d)

When an interrupt occurs, may change state of interrupted process to 'blocked' and schedule another process.

5.5 (a)

Hard disk should be higher priority compared to printer, keyboard, and floppy disk. Because all user programs and OS stored at hard disk.

5.6 (a, c)

(a) When interrupt is enabled, a CPU will be able to process the interrupt.

(b) CPU checks for interrupts before executing a new instruction.

5.7 (c)

Thread → CPU

Virtual address space → Memory

File system → Disk

Signal → Interrupt

5.8 (b)

Shortest seek time next or nearest cylinder next gives the best throughput.

5.9 (d)

Disk requires a device driver in computer system.

5.10 (c)

$$\frac{\text{DB size}}{\text{DBA}} = \text{Number of DBA's possible in one disk block.}$$
5.11 (a)

In a fixed block size file system, larger block size gives better disk throughput but poorer disk space utilization.

5.12 (c)

The number of disk block pointers that will be fit in one block

$$= \frac{2^{10} \text{ byte}}{32 \text{ bit}} = \frac{2^{10} \text{ byte}}{4 \text{ byte}} = 256$$

Maximum file size due to single indirection pointer

$$= 256 \times 1 = 256 \text{ KB}$$

Maximum file size due to direct pointer

$$= 10 \times 1 = 10 \text{ KB}$$

Maximum file size due to double indirection pointer

$$= 256 \times 256 \times 1 \text{ KB}$$

Maximum file size due to triple indirection pointer

$$= 256 \times 256 \times 256 \times 1 \text{ KB}$$

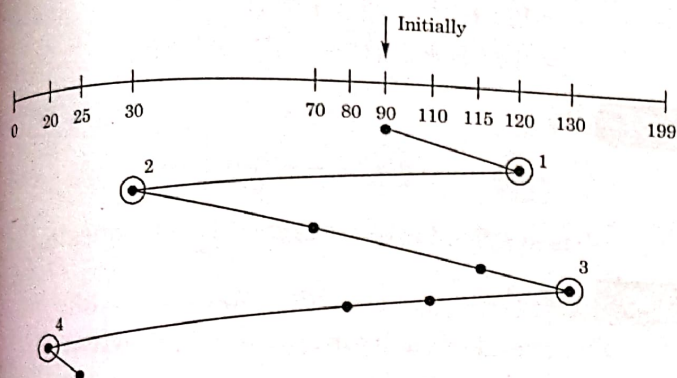
So the maximum file size

$$= 256 \times 256 \times 256 \times 1 \text{ KB}$$

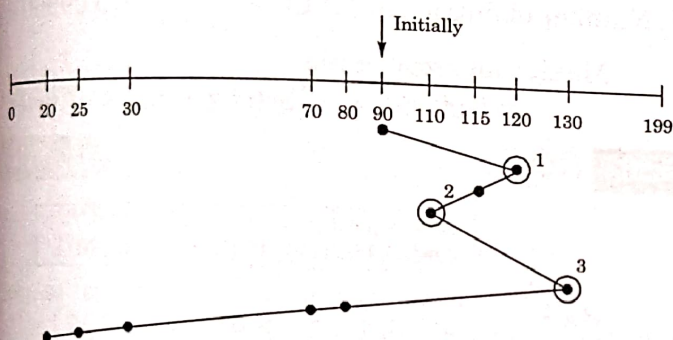
$$= 2^8 \times 2^8 \times 2^8 \times 2^{10} \text{ bytes}$$

$$= 2^{24} \times 2^{10} \text{ bytes} = 2^{34} \text{ bytes}$$

5.13 (c)
Using FCFS:



So, 4 head direction change.
Using SRTF:



So, 3 head direction changed.

5.14 (b)

Maximum file size = [direct DBA + Number of (Data block size/DBA) + Number of (Data block size/DBA)² + Number of (Data block size/DBA)³] * Data block size.

Data Block size = 1024 byte

(Disk Block size/DBA) = Number of Disk Block address Store inside One Block.

Maximum file size = [10 + 1*(128) + 1*(128*128) + 1*(128*128*128)] * 1024 Byte
= Approx 2 GB.

5.15 (c)

Using 100 bytes disk block:

100 B

File size	100 B block	Bytes needed for Book keeping	Block for Information
11050	111	111 × 4 = 444	5
4990	50	50 × 4 = 200	2
5170	52	52 × 4 = 208	3
12640	127	127 × 4 = 508	6
Total	340		16

$$\text{Space} = (340 + 16) \times 100 = 35600$$

Using 200 bytes disk block:

200 B

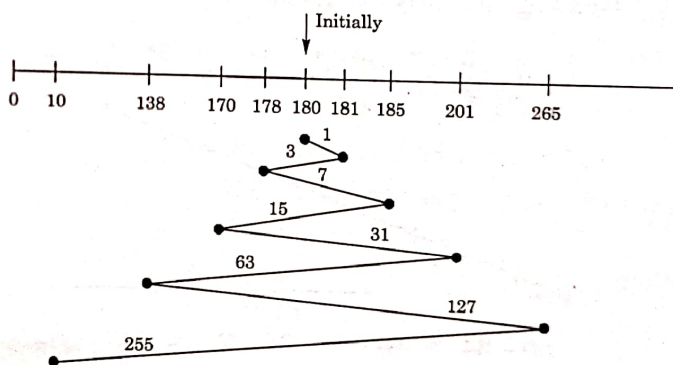
File size	200 B block	Bytes needed for Book keeping	Block for Information
11050	56	56 × 4 = 224	2
4990	25	25 × 4 = 100	1
5170	26	26 × 4 = 104	1
12640	64	64 × 4 = 256	2
Total	171		6

$$\text{Space} = (171 + 6) \times 200 = 35400$$

5.16 (b)

If there are enough extra frames, another process can be initiated. If the sum of the working-set sizes increases, exceeding the total number of available frames, the operating system selects a process to suspend. The process's pages are written out (swapped), and its frames are reallocated to other processes. The suspended process can be restarted later."

5.17 (b)



Initially at 180, so need to go at 181 since all option contain it.

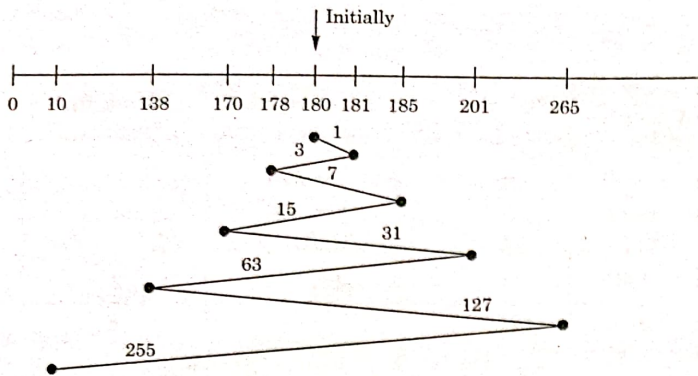
Now 181 - 184 = 3 and 181 - 178 = 3 and 181 - 185 = 4.

Since on tie direction is not changed. So request service next will be 185. So option (b) and (d) can be answer.

Now after services 185, 185 - 200 = 15 and 185 - 170 = 15 and 185 - 201 = 16. So we need to choose request 201 since with 200 direction will not change.

5.18 (c)

If we see the pattern of distance of request, then 1, 3, 7, 15, 31, 63, 127, 255, 511, 1023, 2047 by using, new min distance = (old distance × 2) + 1. So, maximum 11 request can be satisfied.



5.19 (b)

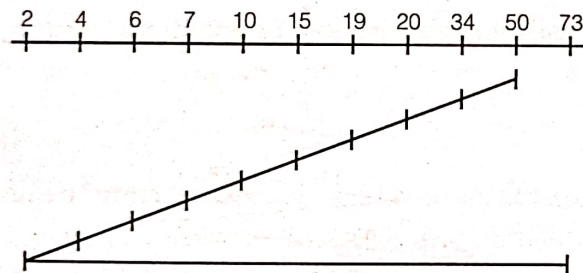
P and Q both are true but Q is not the reason for P because increasing the number of page frames allocated to a process sometimes increases the page fault rate or it is not concern with replacement policy.

5.20 (d)

An extension of index allocation is used to allocate the data block of a very large file.

5.21 (b)

Head is currently at cylinder 50,



$$50 - 34 = 16, \quad 34 - 20 = 14$$

$$20 - 19 = 1, \quad 19 - 15 = 4$$

$$15 - 10 = 5, \quad 10 - 7 = 3$$

$$7 - 6 = 1, \quad 6 - 4 = 2$$

$$4 - 2 = 2, \quad 2 - 73 = 71$$

$$\text{Total moves} = 16 + 14 + 1 + 4 + 5 + 3 + 1 + 2 + 2 + 71 = 119$$

5.22 (b)

Multilevel page table is preferred to reduce the size of page table needed to implement the virtual address space of a process.

5.23 (b)

Maximum possible size

$$= \left[\left(\text{address pointed by doubly indirect block} \right)^2 + \left(\text{address pointed by single address} \right) \right]$$

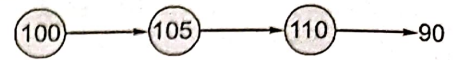
$$+ \text{Address points by single direct address} \times [\text{Block size}]$$

$$= \left[\left(\frac{128}{8} \right)^2 + \left(\frac{128}{8} \right) + 8 \right] \times 128 \text{ B}$$

$$= [2^8 + 2^4 + 2^3] \times 128 \text{ B}$$

$$= 32 \text{ kB} + 2 \text{ kB} + 1 \text{ kB} = 35 \text{ kB}$$

5.24 (3)



90 is serviced after servicing the 3 requests.

5.25 (99.6)

Overhead of each entry in FAT = 4 bytes

Block size = 10^3 bytes

Total size for each entry = 1004 bytes

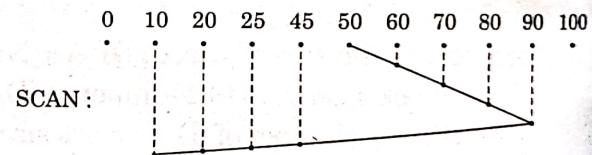
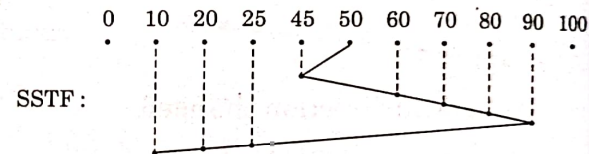
$$\text{Number of entries in FAT} = \frac{100 \times 10^6}{1004} = 0.099601$$

Maximum size of a file

$$= 0.099601 \times 10^3 \text{ bytes} = 99.601 \times 10^6 \text{ bytes}$$

5.26 (10)

Disk request sequence track numbers:
(Order) 45, 20, 90, 10, 50, 60, 80, 25, 70



$$\text{SSTF distance} = (50 - 45) + (90 - 45) + (90 - 10)$$

$$= 5 + 45 + 80 = 130$$

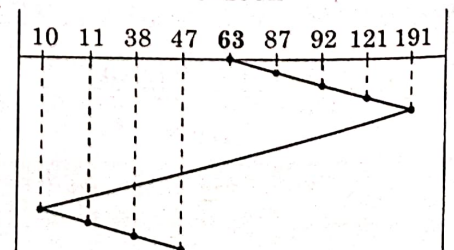
$$\text{SCAN distance} = (90 - 50) + (90 - 10)$$

$$= 40 + 80 = 120$$

$$\therefore \text{SSTF distance} - \text{SCAN distance} = 130 - 120 = 10$$

5.27 (346)

C-Look



$$(84 - 63) + (92 - 87) + (121 - 92) + (191 - 121)$$

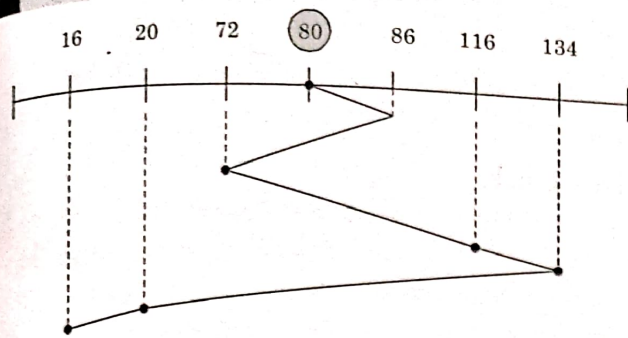
$$+ (191 - 10) + (11 - 10) + (38 - 11) + (47 - 38)$$

$$= 24 + 5 + 29 + 70 + 181 + 1 + 27 + 9 = 346$$

5.28 (d)

Linked and indexed allocations are non contiguous so, they will not suffer from external fragmentation.

5.29 (85)



$$(86 - 80) + (86 - 72) + (134 - 72) + (134 - 16) + 62 + 118 + 14 = 200$$

$$100 \rightarrow 20$$

$$200 \rightarrow ?$$

$$\frac{200}{100} \times 20 = 40$$

$$3 \text{ direction changes } 3 \times 15 = 45$$

$$40 + 45 = 85$$

5.30 (4)

Size of disk block = 4 kB

Disk block address = 32 bit = 4 B

$$\text{Number of address in one disk block} = \frac{4 \text{ kB}}{4 \text{ B}} = 2^{10}$$

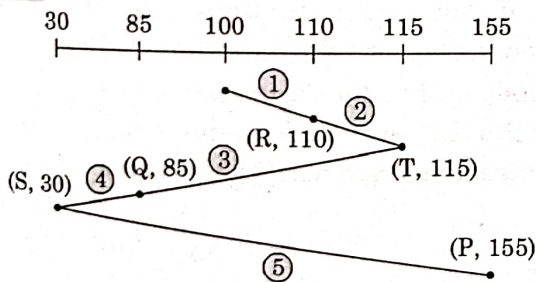
12 direct, 1 single indirect, 1 double indirect pointers.

Maximum possible file size

$$= (12 + 2^{10} + 2^{10} \times 2^{10}) \times 4 \text{ kB}$$

$$= 4.00395 \text{ GB} \approx 4 \text{ GB}$$

5.31 (c)



- Option (a) is correct as P is serviced last.
- Option (b) is correct as P is serviced last.
- Option (c) is false as Q is serviced before S.
- Option (d) is correct.

5.33 (153)

Contiguous Allocation: 50 read operations + 50 write operations + 1 operation to write the newly inserted block = 101 operations (n_A)

Linked Allocation: 50 read operations + 1 operation delete next pointer of 50th element + 1 operation to connect it to the 51st element = 52 operations (n_B)

Total operations:

$$\text{Required} = n_A + n_B = 101 + 52 = 153$$

5.34 (4096)

Disk capacity = Number of surfaces $\times$ Number of tracks per surface $\times$ Number sectors per track $\times$ Number of bytes per sector

$$512 \text{ GB} = 32 \times x \times 4096 \times 1024 \text{ B}$$

$$2^{39} \text{ B} = 2^5 \times x \times 2^{12} \times 2^{10} \text{ B}$$

$$2^{39} \text{ B} = 2^{27} x \text{ B}$$

$$x = \frac{2^{39} \text{ B}}{2^{27} \text{ B}} = 2^{12} = 4096$$

5.35 (30.06)

Average seek time = 5 ms

6000 revolutions 1 min (60 sec)

1 revolution ?

$$\Rightarrow \left(\frac{60}{6000} \right) \text{ sec} \Rightarrow 10 \text{ ms}$$

$$\text{Average Rotational latency} = \frac{1}{2} \times 10 \text{ ms} = 5 \text{ ms}$$

1 revolution time 500 sectors

? 1 sector

$$= \frac{10 \text{ ms}}{500} = 0.02 \text{ ms}$$

$$T_{\text{avg}} = \text{Average seek time} + \text{Average Rotational latency} + \text{Transfer time} + \text{Over head}$$

$$T_{\text{avg}} = (5 \text{ ms} + 5 \text{ ms} + 0.02 \text{ ms} + 0) 3000 = 30060 \text{ ms} = 30.06 \text{ sec}$$

5.36 (65468)

Disk size = 512 MB

$$\text{Block size} = 64 \text{ K} = 2^{16} = 65536$$

Pointer size/block = 4 B, File size = 1 MB

$$\text{Actual space} = 65536 - 4 = 65532$$

$$\text{Number of blocks} = \frac{\text{Total file size}}{\text{Usable data per block}}$$

$$= \frac{1 \text{ MB}}{65532} = 16.007$$

So, data stored in the firm 16 block

$$16 \times 65532 = 1048512 \text{ B}$$

Remaining data still need to store

$$1048756 - 1048512 = 244 \text{ B}$$

Last block contain 64 bytes of the actual data

$$\text{In last block used data} = 64 + 4 = 68 \text{ bytes}$$

Waste space in the last block

$$= 65536 - 18 = 65468 \text{ bytes}$$

5.37 (16417)

Disk size = 16 GB

Block size = 2 KB

Block number = 32 bit = 4 Bytes

Pointer per block : 4 bytes (used to point to next free block)

$$\text{Total number of disk blocks} = \frac{\text{Disk size}}{\text{Block size}}$$

$$= \frac{16 \text{ GB}}{2 \text{ KB}} = \frac{2^4 \times 2^{30}}{2 \times 2^{10}} = \frac{2^{34}}{2^{11}} = 2^{23} = 8388608$$

Each free list block store block numbers + 1 pointer

Usable space for block number per block = 2048 - 4 = 2044 bites

$$\text{Block number usable per block} = \frac{2044}{4} = 511$$

$$\text{Total free list blocks} = \left[\frac{\text{Total disk blocks}}{\text{Block number usable per block}} \right]$$

$$= \left[\frac{8388608}{511} \right] = 16417$$